

DATA STRUCTURE

LAB PROGRAMS

5. Write a C program to find Factorial of a given number using Recursion

```
#include <stdio.h>
```

```
long long factorial(int n) {  
    if(n == 0 || n == 1)  
        return 1;  
    return n * factorial(n - 1);  
}  
  
int main() {  
    int n;  
    printf("Enter a number: ");  
    scanf("%d", &n);  
    printf("Factorial = %lld", factorial(n));  
    return 0;  
}
```

Output

```
Enter a number: 5
```

```
Factorial = 120
```

```
=== Code Execution Successful ===
```